Efficiency–TFP Analysis Kit v3.0

README & User Guide

MAPS Project — Deliverable 4.2 (WP4)

Horizon Europe — Grant Agreement No. 101137914

1. Overview

The **Efficiency–TFP Analysis Kit (v3.0)** is an R script that tests whether aggregate Total Factor Productivity (TFP) can be statistically explained by changes in final-to-useful exergy efficiency, following the econometric methodology of De Ketelaere et al. (2026) and Santos et al. (2021). For any country for which the required macroeconomic and exergy data are available, the kit runs a complete eight-step cointegration analysis pipeline, writes all results to a formatted multi-sheet Excel workbook, and generates a structured PDF report.

The kit is part of the MAPS Integrated Assessment Modelling project and is designed for use by researchers familiar with time-series econometrics. No manual coding is required beyond a single function call.

2. Software Requirements

R version 4.1 or later. All R package dependencies are installed automatically on first run if not already present.

Package	Purpose
`openxlsx`	Excel output (xlsx)
`urca`	Johansen cointegration (ca.jo)
`vars`	VAR estimation and diagnostics
`ggplot2`	Charts in the PDF report
`gridExtra`	Multi-panel chart layouts
`grid`	Low-level PDF page composition
`scales`	Axis and number formatting
`tools`	File-path utilities (base R)

3. Loading the Kit

Place eff_tfp_v3.R in your R working directory and source it once per session:

```
source("eff_tfp_v3.R")
```

This makes two functions available: run_analysis() for any country with a data file, and run_portugal_santos2021() for the built-in Portugal dataset.

4. Running the Analysis

4.1 Standard country analysis

Call run_analysis() with the path to your Excel or CSV data file. The function runs the full pipeline and writes both output files.

```
# Minimal call — uses all defaults
results <- run_analysis("austria.xlsx")

# With optional arguments
results <- run_analysis(
    file_path = "austria.xlsx",
    var_lag   = 2,                           # VAR lag order (default:
BIC selection)
    out_path  = "my_results.xlsx"           # custom Excel output name
)
```

Output files are saved in the working directory with names derived automatically from the input file (see Section 8).

4.2 Portugal — Santos et al. (2021)

The kit includes a **built-in pre-loaded dataset** for **Portugal, 1960–2014**, based on Santos et al. (2021). This covers the TFP series (unadjusted and quality-adjusted), the final-to-useful exergy efficiency series, the useful exergy intensity, the output elasticities, and the supporting diagnostics from the original paper. Call the dedicated function to run the full analysis pipeline on this dataset:

```
# Run with default output names
run_portugal_santos2021()

# With custom output file names
run_portugal_santos2021(
    out_excel = "portugal_santos.xlsx",
    out_pdf   = "portugal_santos.pdf"
)
```

The function runs the same eight-step analytical pipeline as any other country. It produces an identical Excel workbook and PDF report structure, with results reflecting the cointegration analysis applied to the Santos et al. (2021) dataset. Because this dataset does not include the underlying capital stock, labour, and GDP time series, the kit uses the **pre-loaded TFP and efficiency series as its starting point** (equivalent to using the pre-loaded data template described in Section 5.2), and the Cobb-Douglas GDP estimation uses the pre-computed GDP series embedded alongside the TFP data.

> **Note:** The Portugal Santos 2021 dataset is a validated reference case. Comparing your country results against it is a useful consistency check.

5. Input Data Format

5.1 Standard template — full macro and exergy data

Prepare an Excel file using the layout of Sheet 1_Standard_Template in data_template_v3.xlsx. The header row is row 21 and data begin at row 22. Columns below marked * are required; others are used when available and skipped otherwise.

Column	Description	Unit / Notes
year *	Calendar year	Integer
gdp *	Gross domestic product	M 2021 USD PPP

Column	Description	Unit / Notes
k_stock	Net capital stock (unadj.)	M 2021 USD
k_services	Capital services (quality-adj.)	M 2021 USD
emp *	Employed persons	Thousands
avh	Average annual hours worked	Hours/person
hc	Human capital index	Index
adj_gos *	Adjusted gross operating surplus	M 2021 USD
gos *	Gross operating surplus	M 2021 USD
coe *	Compensation of employees	M 2021 USD
x_final *	Final exergy	PJ
x_useful *	Useful exergy	PJ

5.2 Pre-loaded template — TFP and efficiency series only

If you already have computed, normalised TFP and efficiency series (e.g. from a prior run or external source), use the pre-loaded template format (Sheet 2_Preloaded_Template in data_template_v3.xlsx). The kit detects this format automatically and skips Steps 2–5 of the pipeline.

Column	Description
year	Calendar year (integer)
tfp_unadj	Unadjusted TFP — z-normalised or raw Solow residual
tfp_adj	Quality-adjusted TFP (optional)
eff	Final-to-useful exergy efficiency (ratio or percentage)

Note: The Portugal Santos 2021 built-in dataset uses the pre-loaded format internally, since only the TFP and efficiency series — not the underlying factor inputs — are available from the published paper.

6. Analysis Pipeline

The kit runs eight sequential steps. Steps 2–5 are skipped when the input is in pre-loaded format (Section 5.2).

Step 1 — Output elasticities (factor shares)

The capital output elasticity is computed annually as **α_K = adj_GOS / (GOS + COE)**. The labour elasticity is $\alpha\_L = 1 - \alpha\_K$. The full-sample time-average α_K is used as a constant for TFP decomposition and Cobb-Douglas GDP estimation. If income-share data are absent, the kit falls back to the standard assumed values $\alpha\_K = 0.30$, $\alpha\_L = 0.70$.

Step 2 — Labour inputs

Two labour aggregates are constructed: unadjusted labour **L = emp × avh**, and quality-adjusted labour **hL = emp × avh × hc**. If avh is unavailable, emp is used directly. If hc is absent, hL = L.

Step 3 — TFP series (Solow residual)

TFP is computed as the Solow residual: **TFP = GDP / (K^α_K × L^α_L)**. Both unadjusted (K_stock, L) and quality-adjusted (K_services, hL) variants are produced. All series are z-normalised to the base year: **z_x = ln(x / x_0) + 1**, so every series equals 1.000 at the first observation.

Step 4 — Exergy efficiency and intensity

Final-to-useful exergy efficiency: ε = X_useful / X_final. Useful exergy intensity: X_useful / GDP. The z-normalised efficiency series z_EFF enters the cointegration analysis; both raw ε values and intensity values are stored in the output.

Step 5 — Unit root tests (ADF and PP)

Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests are applied to z_TFP_unadj, z_TFP_adj, and z_EFF in levels (with trend and intercept) and first differences (with intercept). ADF lag length is selected by BIC on a fixed-sample basis following EViews conventions. PP bandwidth is Newey-West. Critical values follow MacKinnon (1996).

Step 6 — VAR estimation and residual diagnostics

A bivariate VAR(p) is estimated for each TFP–EFF pair. Three residual diagnostic tests are applied:

- Portmanteau (cumulative Q-statistic up to lag h, valid for h > p)
- LM serial correlation (Edgeworth-corrected LR statistic and Rao F-statistic, both at individual lags and cumulatively from lag 1 to h)
- Jarque-Bera normality (Cholesky decomposition of residual covariance, Lütkepohl correction, by equation and jointly)

Step 7 — Johansen cointegration

Two deterministic specifications are tested:

- Hz — no deterministic term in the cointegrating equation
- Hc — restricted constant in the cointegrating equation (intercept enters only through the CE, not the short-run dynamics)

Both Trace and Max-Eigenvalue statistics are reported. P-values follow MacKinnon-Haug-Michelis (1999). The normalised cointegrating vector **β** (the EFF elasticity of TFP) and its standard error are estimated via concentrated OLS on the manually constructed Johansen R0/R1 residual matrices. The preferred specification is quality-adjusted TFP under Hc.

Step 8 — Useful exergy intensity constancy

The useful exergy intensity series is tested for constancy over (a) the full sample and (b) the last third of the sample. The constancy criterion is: **constant if (trend p-value > 0.10 AND at least one of ADF or PP confirms I(0)) OR CV < 5%**. Descriptive statistics, OLS trend results, and ADF/PP test statistics are reported for both sub-periods.

7. Output Files

7.1 Excel workbook

The workbook (eff_tfp_results_[country].xlsx) contains nine sheets, each with a colour-coded header row and alternating-row shading:

Sheet	Contents
1_Elasticities	Annual α_K and α_L; cumulative means
2_TFP_Series	z_TFP_unadj, z_TFP_adj, z_EFF, raw ε (%), intensity (X_useful/GDP)

Sheet	Contents
3_Unit_Root_Tests	ADF and PP: statistic, critical values, auxiliary parameter, verdict
4_VAR_Diagnostics	Portmanteau, LM individual and cumulative, JB normality
5_Johansen_Hz	Trace and Max-Eigenvalue tables, normalised CE, standard errors (Hz)
6_Johansen_Hc	Trace and Max-Eigenvalue tables, normalised CE, standard errors (Hc)
7_Intensity_Constancy	Descriptive stats, trend test, ADF/PP results, constancy verdict (full + sub)
8_Estimated_TFP	Historical vs fitted TFP for Hz and Hc; RMSE, MAE, R^2, Corr, MAPE
9_Estimated_GDP	Historical vs fitted GDP (Cobb-Douglas APF); KL component; goodness-of-fit

7.2 PDF report

The PDF report (eff_tfp_report_[country].pdf) is an A4 document with a blue header bar on every page showing the country name, section title, kit version, and page number. It contains:

Page(s)	Contents
Cover	Title, country, period, MAPS reference; opening paragraph; steps summary; key results: CV equation, Cobb-Douglas APF, intensity verdict
1	Annual output elasticities (α_K, α_L) with full-sample mean lines and fixed reference lines at 0.30/0.70
2	Normalised z_TFP_unadj, z_TFP_adj, and z_EFF series on a single time-series chart
3	ADF and PP unit root test tables for all three series, with verdict banner
4	Portmanteau residual autocorrelation test table; [OK]/[!!] verdict banner
5	LM serial correlation test (individual and cumulative lags); verdict banner
6	Jarque-Bera normality test (by equation and joint); verdict banner
7	Johansen cointegration — Hz specification: Trace, Max-Eigenvalue, normalised CE
8	Johansen cointegration — Hc specification: Trace, Max-Eigenvalue, normalised CE
9	Fitted vs historical TFP — Hz specification (unadjusted and quality-adjusted panels)
10	Fitted vs historical TFP — Hc specification (unadjusted and quality-adjusted panels)
11	Fitted vs historical GDP — Cobb-Douglas APF with KL factor accumulation component
12	Useful exergy intensity: normalised series, reference/trend line, constancy table
13	References and funding acknowledgement

8. Output File Naming

Output file names are derived automatically from the input file name (extension stripped, title-cased). Custom names can be specified via the out_path argument.

Input / call	Excel output	PDF report
`austria.xlsx`	`eff_tfp_results_austria.xlsx`	`eff_tfp_report_austria.p`
`fra_2023.xlsx`	`eff_tfp_results_fra_2023.xlsx`	`eff_tfp_report_fra_2023.`
`run_portugal_santos2021()`	`eff_tfp_results_portugal_santos2021.xlsx`	`eff_tfp_report_portugal_`

9. Technical Notes

- z-normalisation $z\_x = \ln(x/x_0) + 1$ ensures all series equal 1.000 at the base year and are comparable across countries without rescaling. The cointegrating relationship in z-form reads $z\_TFP = \beta \times z\_EFF$ (Hz) or $z\_TFP = \beta \times z\_EFF + C$ (Hc), which translates to approximately $TFP = \exp(-C) \times EFF^{\beta}$ in levels.
- The ADF lag-length BIC uses a fixed sample of size $T - p - 1$ to ensure comparability across lag orders, consistent with EViews convention.
- Johansen standard errors are obtained from concentrated OLS on the manually constructed R0/R1 residual matrices, bypassing urca slot-naming inconsistencies.
- The preferred specification throughout is quality-adjusted TFP (K_services, hL) under the Hc deterministic assumption, as this allows for a constant shift in the long-run TFP–efficiency relationship and uses the most comprehensive measure of productive inputs.
- Intensity constancy uses a two-period assessment (full sample + last third) so that constancy in recent decades can be recognised even if the full historical series is non-stationary.

10. References

De Ketelaere, J., Santos, J., Domingos, T. (2026, under review). Total Factor Productivity is fully explained by changes in physical energy efficiency: an exergy economics approach. Energy Economics.

MacKinnon, J. G. (1996). Numerical distribution functions for unit root and cointegration tests. Journal of Applied Econometrics, 11(6), 601–618.

MacKinnon, J. G., Haug, A. A., Michelis, L. (1999). Numerical distribution functions of likelihood ratio tests for cointegration. Journal of Applied Econometrics, 14(5), 563–577.

Santos, J., Borges, A. S., Domingos, T. (2021). Exploring the links between total factor productivity and energy efficiency: Portugal, 1960–2014. Energy Economics, 101, 105407.